

Event Finance Manager - Database Schema Documentation

Entity Relationship Diagram

Complete ER Diagram

```
erDiagram
%% Multi-Tenancy Layer
Organization ||--o{ User : "has"
Organization ||--o{ Event : "has"
Organization ||--o{ Vendor : "has"
Organization ||--o{ Expense : "has"
Organization ||--|| DashboardMetrics : "has"
Organization ||--o{ Subscription : "has"
Organization ||--o{ Notification : "has"
Organization ||--o{ ActivityLog : "has"
```

```
%% User Relationships
User ||--o{ Event : "creates"
User ||--o{ Expense : "creates"
User ||--o{ EventAssignment : "assigned"
User ||--o{ BudgetItem : "assigned"
User ||--o{ ApprovalWorkflow : "approves"
User ||--o{ Notification : "receives"
User ||--o{ ActivityLog : "performs"
User ||--o{ Report : "creates"
User ||--o{ File : "uploads"
User ||--o{ Note : "creates"
User ||--o{ SubscriptionHistory : "changes"
User ||--o{ Organization : "deletes"
User ||--o{ User : "deletes"
User ||--o{ Event : "deletes"
User ||--o{ BudgetItem : "deletes"
User ||--o{ Expense : "deletes"
User ||--o{ Vendor : "deletes"
User ||--o{ File : "deletes"
User ||--o{ Note : "deletes"
User ||--o{ Report : "deletes"
User ||--o{ StrategicGoal : "deletes"
User ||--o{ EventAssignment : "deletes"
User ||--o{ EventStakeholder : "deletes"
User ||--o{ VendorEvent : "deletes"
User ||--o{ Insight : "deletes"
User ||--o{ AiBudgetSuggestion : "deletes"
User ||--o{ Subscription : "deletes"

%% Event Relationships
Event ||--o{ EventAssignment : "has"
Event ||--o{ EventStakeholder : "has"
Event ||--o{ BudgetItem : "has"
Event ||--o{ Expense : "has"
Event ||--o{ VendorEvent : "has"
```

```

Event ||--o{ StrategicGoal : "has"
Event ||--|| EventMetrics : "has"
Event ||--|| ROIMetrics : "has"
Event ||--|| CRMSync : "has"
Event ||--o{ File : "has"
Event ||--o{ Note : "has"
Event ||--o{ Insight : "has"
Event ||--o{ Report : "has"
Event ||--o{ AiBudgetSuggestion : "has"
Event ||--o{ ActivityLog : "has"

```

%% Budget Relationships

```

BudgetItem }o--|| Event : "belongs to"
BudgetItem }o--o| Vendor : "uses"
BudgetItem }o--o| User : "assigned to"
BudgetItem }o--o| StrategicGoal : "linked to"
BudgetItem ||--o{ Expense : "tracks"
BudgetItem ||--o{ ApprovalWorkflow : "has"
BudgetItem ||--o{ File : "has"

```

%% Expense Relationships

```

Expense }o--|| Event : "belongs to"
Expense }o--o| BudgetItem : "linked to"
Expense }o--o| Vendor : "uses"
Expense }o--|| User : "created by"
Expense ||--o{ ApprovalWorkflow : "has"
Expense ||--o{ File : "has"

```

%% Vendor Relationships

```

Vendor ||--o{ VendorEvent : "assigned to"
Vendor ||--o{ BudgetItem : "used in"
Vendor ||--o{ Expense : "used in"
Vendor ||--|| VendorMetrics : "has"

```

%% Approval Workflow

```

ApprovalWorkflow }o--o| Expense : "tracks"
ApprovalWorkflow }o--o| BudgetItem : "tracks"
ApprovalWorkflow }o--|| User : "by"

```

%% Subscription

```

Subscription ||--o{ SubscriptionHistory : "has"
SubscriptionHistory }o--o| User : "changed by"

```

%% File Relationships

```

File }o--o| Event : "attached to"
File }o--o| BudgetItem : "attached to"
File }o--o| Expense : "attached to"
File }o--o| Report : "attached to"
File }o--|| User : "uploaded by"

```

%% Report Relationships

```
Report }o--|| Event : "for"
Report }o--|| User : "created by"
Report ||--o{ File : "has"

%% Entity Definitions with Soft Delete Fields
Organization {
    string id PK
    string name
    string baseCurrency
    datetime deletedAt "Soft Delete"
    string deletedBy FK "Soft Delete"
    datetime createdAt
    datetime updatedAt
}

User {
    string id PK
    string organizationId FK
    string fullName
    string email UK
    enum role
    boolean isActive
    boolean emailVerified
    datetime deletedAt "Soft Delete"
    string deletedBy FK "Soft Delete"
    datetime createdAt
    datetime updatedAt
}

Event {
    string id PK
    string organizationId FK
    string name
    string eventType
    enum status
    decimal budget
    string currency "PLAN Module"
    string businessUnit "PLAN Module"
    string eventObjective "PLAN Module"
    int internalAttendees "PLAN Module"
    int externalAttendees "PLAN Module"
    datetime deletedAt "Soft Delete"
    string deletedBy FK "Soft Delete"
    string createdBy FK
    datetime createdAt
    datetime updatedAt
}

BudgetItem {
    string id PK
    string eventId FK
```

```
enum category
string subcategory
decimal estimatedCost
decimal actualCost
decimal allocatedAmount
enum status
datetime deletedAt "Soft Delete"
string deletedBy FK "Soft Delete"
datetime createdAt
datetime updatedAt
}
```

```
Expense {
  string id PK
  string organizationId FK
  string eventId FK
  string budgetItemId FK
  string vendorId FK
  string title
  float amount
  enum status
  datetime deletedAt "Soft Delete"
  string deletedBy FK "Soft Delete"
  string createdBy FK
  datetime createdAt
  datetime updatedAt
}
```

```
Vendor {
  string id PK
  string organizationId FK
  string name
  string email
  float rating
  datetime deletedAt "Soft Delete"
  string deletedBy FK "Soft Delete"
  datetime createdAt
  datetime updatedAt
}
```

```
ApprovalWorkflow {
  string id PK
  string expenseId FK
  string budgetItemId FK
  string approverId FK
  string action
  string comments
  datetime actionAt
}
```

```
File {
```

```
    string id PK
    string eventId FK
    string budgetItemId FK
    string expenseId FK
    string filename
    string path
    datetime deletedAt "Soft Delete"
    string deletedBy FK "Soft Delete"
    datetime uploadedAt
}
```

```
Note {
    string id PK
    string eventId FK
    string content
    string_array tags
    datetime deletedAt "Soft Delete"
    string deletedBy FK "Soft Delete"
    datetime createdAt
}
```

```
Report {
    string id PK
    string eventId FK
    string reportType
    datetime deletedAt "Soft Delete"
    string deletedBy FK "Soft Delete"
    datetime createdAt
}
```

```
StrategicGoal {
    string id PK
    string eventId FK
    string title
    float targetValue
    datetime deletedAt "Soft Delete"
    string deletedBy FK "Soft Delete"
    datetime createdAt
}
```

```
EventAssignment {
    string id PK
    string userId FK
    string eventId FK
    string role
    datetime deletedAt "Soft Delete"
    string deletedBy FK "Soft Delete"
    datetime assignedAt
}
```

```
EventStakeholder {
```

```
    string id PK
    string eventId FK
    string name
    string email
    datetime deletedAt "Soft Delete"
    string deletedBy FK "Soft Delete"
    datetime createdAt
}

VendorEvent {
    string id PK
    string vendorId FK
    string eventId FK
    datetime deletedAt "Soft Delete"
    string deletedBy FK "Soft Delete"
    datetime assignedAt
}

Insight {
    string id PK
    string eventId FK
    string insightType
    json data
    datetime deletedAt "Soft Delete"
    string deletedBy FK "Soft Delete"
    datetime createdAt
}

AiBudgetSuggestion {
    string id PK
    string eventId FK
    enum category
    decimal suggestedCost
    boolean accepted
    datetime deletedAt "Soft Delete"
    string deletedBy FK "Soft Delete"
    datetime createdAt
}

Subscription {
    string id PK
    string organizationId FK
    string planName
    enum status
    datetime deletedAt "Soft Delete"
    string deletedBy FK "Soft Delete"
    datetime createdAt
}

DashboardMetrics {
    string id PK
```

```
    string organizationId FK UK
    decimal totalBudget
    decimal totalExpenses
    int pendingApprovals
    datetime lastComputedAt
}

EventMetrics {
    string id PK
    string eventId FK UK
    decimal totalBudget
    decimal totalSpent
    decimal variance
    boolean isOverBudget
    datetime lastComputedAt
}

VendorMetrics {
    string id PK
    string vendorId FK UK
    int totalContracts
    decimal totalSpent
    datetime lastComputedAt
}---
```

Database Tables Overview

Core Tables

1. Organization

Purpose: Multi-tenancy root table. Each organization is a separate tenant.

Key Fields:

- **id** (PK): Unique identifier
- **name**: Organization name
- **baseCurrency**: Default currency (default: USD)
- **deletedAt**: Soft delete timestamp
- **deletedBy**: User who deleted the organization

Relationships:

- Has many Users
- Has many Events
- Has many Vendors
- Has many Expenses
- Has one DashboardMetrics

2. User

Purpose: System users with role-based access control.

Key Fields:

- **id** (PK): Unique identifier
- **email** (UK): Unique email address
- **role**: UserRole enum (Admin, EventManager, Finance, Viewer)
- **organizationId** (FK): Belongs to Organization
- **isActive**: Account status
- **emailVerified**: Email verification status
- **deletedAt**: Soft delete timestamp
- **deletedBy**: User who deleted this user

Relationships:

- Belongs to Organization
 - Creates Events
 - Creates Expenses
 - Assigned to Events (via EventAssignment)
 - Assigned to BudgetItems
 - Approves workflows
 - Uploads Files
 - Creates Notes and Reports
-

3. Event

Purpose: Core event entity with PLAN module fields.

Key Fields:

- **id** (PK): Unique identifier
- **name**: Event name
- **eventType**: Virtual, In-Person, Hybrid, City-wide
- **status**: EventStatus enum (Planning, Active, Completed, Cancelled)
- **currency**: Event currency (default: USD)
- **budget**: Total budget amount
- **businessUnit**: Business unit or department
- **eventObjective**: Lead Generation, Customer Engagement, etc.
- **internalAttendees**: Internal attendee count
- **externalAttendees**: External attendee count
- **deletedAt**: Soft delete timestamp
- **deletedBy**: User who deleted the event

Relationships:

- Belongs to Organization
- Created by User

- Has many BudgetItems
 - Has many Expenses
 - Has many EventAssignments
 - Has many EventStakeholders
 - Has many VendorEvents
 - Has one EventMetrics
 - Has one ROIMetrics
 - Has one CRMSync
-

4. BudgetItem

Purpose: Budget line items for events.

Key Fields:

- **id** (PK): Unique identifier
- **eventId** (FK): Belongs to Event
- **category**: BudgetItemCategory enum
- **subcategory**: Subcategory name
- **estimatedCost**: Estimated amount
- **actualCost**: Actual amount spent
- **allocatedAmount**: Allocated amount
- **status**: BudgetItemStatus enum (Pending, Approved, Rejected)
- **vendorId** (FK): Optional vendor link
- **assignedUserId** (FK): Assigned user
- **deletedAt**: Soft delete timestamp
- **deletedBy**: User who deleted the item

Relationships:

- Belongs to Event
 - Links to Vendor (optional)
 - Assigned to User (optional)
 - Links to StrategicGoal (optional)
 - Tracks Expenses
 - Has ApprovalWorkflows
 - Has Files
-

5. Expense

Purpose: Actual expenses incurred for events.

Key Fields:

- **id** (PK): Unique identifier
- **eventId** (FK): Belongs to Event
- **budgetItemId** (FK): Optional link to BudgetItem

- **vendorId** (FK): Optional vendor link
- **title**: Expense title
- **amount**: Expense amount
- **status**: ExpenseStatus enum (Pending, Approved, Rejected)
- **createdBy** (FK): User who created
- **deletedAt**: Soft delete timestamp
- **deletedBy**: User who deleted the expense

Relationships:

- Belongs to Event
 - Links to BudgetItem (optional)
 - Links to Vendor (optional)
 - Created by User
 - Has ApprovalWorkflows
 - Has Files (receipts)
-

6. Vendor

Purpose: Vendor/supplier management.

Key Fields:

- **id** (PK): Unique identifier
- **organizationId** (FK): Belongs to Organization
- **name**: Vendor name
- **email**: Contact email
- **serviceType**: Type of service
- **rating**: Vendor rating
- **deletedAt**: Soft delete timestamp
- **deletedBy**: User who deleted the vendor

Relationships:

- Belongs to Organization
 - Used in BudgetItems
 - Used in Expenses
 - Assigned to Events (via VendorEvent)
 - Has VendorMetrics
-

Supporting Tables

7. ApprovalWorkflow

Purpose: Tracks approval/rejection actions for expenses and budget items.

Key Fields:

- **id** (PK): Unique identifier
 - **expenseId** (FK): Optional expense link
 - **budgetItemId** (FK): Optional budget item link
 - **approverId** (FK): User who approved/rejected
 - **action**: "approved" or "rejected"
 - **comments**: Approval comments
 - **actionAt**: Timestamp of action
-

8. File

Purpose: File storage for receipts, documents, attachments.

Key Fields:

- **id** (PK): Unique identifier
 - **eventId** (FK): Optional event link
 - **budgetItemId** (FK): Optional budget item link
 - **expenseId** (FK): Optional expense link
 - **reportId** (FK): Optional report link
 - **filename**: File name
 - **path**: Storage path
 - **mimeType**: File MIME type
 - **size**: File size in bytes
 - **uploadedBy** (FK): User who uploaded
 - **deletedAt**: Soft delete timestamp
 - **deletedBy**: User who deleted the file
-

9. EventAssignment

Purpose: Links users to events with roles.

Key Fields:

- **id** (PK): Unique identifier
 - **userId** (FK): User assigned
 - **eventId** (FK): Event assignment
 - **role**: Assignment role (Manager, Lead, etc.)
 - **deletedAt**: Soft delete timestamp
 - **deletedBy**: User who deleted the assignment
-

10. EventStakeholder

Purpose: External participants/stakeholders for events.

Key Fields:

- **id** (PK): Unique identifier

- **eventId** (FK): Belongs to Event
 - **name**: Stakeholder name
 - **email**: Contact email
 - **role**: Stakeholder role
 - **deletedAt**: Soft delete timestamp
 - **deletedBy**: User who deleted the stakeholder
-

11. VendorEvent

Purpose: Links vendors to events.

Key Fields:

- **id** (PK): Unique identifier
 - **vendorId** (FK): Vendor
 - **eventId** (FK): Event
 - **assignedAt**: Assignment timestamp
 - **deletedAt**: Soft delete timestamp
 - **deletedBy**: User who deleted the assignment
-

12. StrategicGoal

Purpose: Strategic goals linked to events.

Key Fields:

- **id** (PK): Unique identifier
 - **eventId** (FK): Belongs to Event
 - **title**: Goal title
 - **targetValue**: Target value
 - **currentValue**: Current progress
 - **status**: Goal status
 - **priority**: Goal priority
 - **deletedAt**: Soft delete timestamp
 - **deletedBy**: User who deleted the goal
-

13. Note

Purpose: Notes/comments on events.

Key Fields:

- **id** (PK): Unique identifier
- **eventId** (FK): Belongs to Event
- **content**: Note content
- **tags**: Array of tags
- **createdBy** (FK): User who created

- `deletedAt`: Soft delete timestamp
 - `deletedBy`: User who deleted the note
-

14. Report

Purpose: Generated reports for events.

Key Fields:

- `id` (PK): Unique identifier
 - `eventId` (FK): Belongs to Event
 - `reportType`: Type of report
 - `createdBy` (FK): User who created
 - `deletedAt`: Soft delete timestamp
 - `deletedBy`: User who deleted the report
-

15. Subscription

Purpose: Organization subscription management.

Key Fields:

- `id` (PK): Unique identifier
 - `organizationId` (FK): Belongs to Organization
 - `planName`: Subscription plan name
 - `billingCycle`: Monthly or Yearly
 - `status`: SubscriptionStatus enum
 - `deletedAt`: Soft delete timestamp
 - `deletedBy`: User who deleted the subscription
-

16. SubscriptionHistory

Purpose: History of subscription changes.

Key Fields:

- `id` (PK): Unique identifier
 - `subscriptionId` (FK): Belongs to Subscription
 - `action`: Change action (upgrade, downgrade, etc.)
 - `changedBy` (FK): User who made the change
 - `changedAt`: Change timestamp
-

Analytics & Metrics Tables

17. DashboardMetrics

Purpose: Pre-computed organization-level metrics.

Key Fields:

- `id` (PK): Unique identifier
 - `organizationId` (FK, UK): One per organization
 - `totalBudget`: Total budget across all events
 - `totalExpenses`: Total expenses
 - `pendingApprovals`: Count of pending approvals
 - `overBudgetEvents`: Count of over-budget events
 - `lastComputedAt`: Last computation timestamp
-

18. EventMetrics

Purpose: Pre-computed event-level metrics.

Key Fields:

- `id` (PK): Unique identifier
 - `eventId` (FK, UK): One per event
 - `totalBudget`: Sum of budget items
 - `totalSpent`: Sum of approved expenses
 - `variance`: Budget variance
 - `isOverBudget`: Over budget flag
 - `totalsByCategory`: JSON breakdown by category
 - `lastComputedAt`: Last computation timestamp
-

19. VendorMetrics

Purpose: Pre-computed vendor-level metrics.

Key Fields:

- `id` (PK): Unique identifier
 - `vendorId` (FK, UK): One per vendor
 - `totalContracts`: Count of vendor events
 - `totalSpent`: Total amount spent with vendor
 - `lastContractDate`: Last contract date
 - `lastComputedAt`: Last computation timestamp
-

Integration Tables

20. Insight

Purpose: AI-generated insights and recommendations.

Key Fields:

- `id` (PK): Unique identifier
- `eventId` (FK): Belongs to Event

- **insightType**: Type of insight
 - **data**: JSON insight data
 - **deletedAt**: Soft delete timestamp
 - **deletedBy**: User who deleted the insight
-

21. ROIMetrics

Purpose: ROI calculations per event.

Key Fields:

- **id** (PK): Unique identifier
 - **eventId** (FK, UK): One per event
 - **totalBudget**: Total budget
 - **actualSpend**: Actual spend
 - **leadsGenerated**: Number of leads
 - **conversions**: Number of conversions
 - **revenueGenerated**: Revenue amount
 - **roiPercent**: ROI percentage
-

22. CRMSync

Purpose: CRM integration sync status.

Key Fields:

- **id** (PK): Unique identifier
 - **eventId** (FK, UK): One per event
 - **crmSystem**: CRM system name
 - **syncStatus**: Sync status
 - **lastSyncedAt**: Last sync timestamp
 - **data**: JSON sync data
-

23. AiBudgetSuggestion

Purpose: AI-generated budget suggestions.

Key Fields:

- **id** (PK): Unique identifier
- **eventId** (FK): Belongs to Event
- **category**: Budget category
- **suggestedCost**: Suggested cost amount
- **reasoning**: AI reasoning
- **confidence**: Confidence score (0-1)
- **accepted**: Whether suggestion was accepted
- **deletedAt**: Soft delete timestamp

- **deletedBy**: User who deleted the suggestion
-

24. Notification

Purpose: User notifications.

Key Fields:

- **id** (PK): Unique identifier
 - **organizationId** (FK): Optional organization link
 - **userId** (FK): User who receives notification
 - **type**: NotificationType enum
 - **title**: Notification title
 - **message**: Notification message
 - **read**: Read status
 - **readAt**: Read timestamp
-

25. ActivityLog

Purpose: Audit trail of all system actions.

Key Fields:

- **id** (PK): Unique identifier
 - **organizationId** (FK): Optional organization link
 - **userId** (FK): User who performed action
 - **eventId** (FK): Optional event link
 - **action**: Action type (created, updated, deleted)
 - **details**: JSON action details
 - **createdAt**: Action timestamp
-

Soft Delete Implementation

All major tables support soft delete with the following fields:

- **deletedAt**: DateTime? - Timestamp when record was deleted (NULL = not deleted)
- **deletedBy**: String? - User ID who deleted the record (FK to User)

Tables with Soft Delete:

- Organization
- User
- Event
- BudgetItem
- Expense
- Vendor
- File

- Note
- Report
- StrategicGoal
- EventAssignment
- EventStakeholder
- VendorEvent
- Insight
- AiBudgetSuggestion
- Subscription

Tables without Soft Delete (Audit/Logging):

- ApprovalWorkflow (audit trail)
- ActivityLog (audit trail)
- Notification (ephemeral)
- SubscriptionHistory (historical record)
- ROIMetrics (computed)
- CRMSync (sync state)
- DashboardMetrics (computed)
- EventMetrics (computed)
- VendorMetrics (computed)

Relationship Summary

One-to-Many Relationships

- Organization → Users, Events, Vendors, Expenses
- Event → BudgetItems, Expenses, EventAssignments, EventStakeholders
- User → Created Events, Created Expenses, Uploaded Files
- BudgetItem → Expenses, ApprovalWorkflows, Files
- Vendor → BudgetItems, Expenses, VendorEvents

One-to-One Relationships

- Event → EventMetrics, ROIMetrics, CRMSync
- Organization → DashboardMetrics
- Vendor → VendorMetrics

Many-to-Many Relationships

- User ↔ Event (via EventAssignment)
- Vendor ↔ Event (via VendorEvent)
- BudgetItem ↔ StrategicGoal (via strategicGoalId)

Indexes

All tables have appropriate indexes on:

- Primary keys (automatic)
 - Foreign keys
 - Soft delete fields (`deletedAt`, `deletedBy`)
 - Frequently queried fields (status, dates, organizationId)
 - Composite indexes for common query patterns
-

Notes

1. **Soft Delete:** Records are never physically deleted. Use `deletedAt IS NULL` in queries to filter active records.
 2. **Multi-Tenancy:** All data is scoped by `organizationId` for data isolation.
 3. **Audit Trail:** All deletions are tracked via `deletedBy` field linking to User table.
 4. **PLAN Module Fields:** Event table includes fields for event intake, budgeting, and planning workflows.
 5. **Metrics Tables:** Pre-computed metrics are updated via background jobs or triggers for performance.
-

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Schema Version: With Soft Delete Support